

Lec6 Naive Bayes (continued) + Support vector machines

Outline:

Naive Bayes

- Laplace smoothing
- Event models

Comment on apply ML

SVM intros

Recap: Generative model

$$P(x|y) \cdot P(y)$$

$$P(x|y) = \prod_{j=1}^n P(x_j|y)$$

product of the conditional probabilities of the individual features given the class label y

$$\text{parameters: } P(y=1) = \phi_y$$

$$P(x_j=1|y=0) = \phi_{j|y=0} \quad \text{chance of the } j\text{th word appearing in non-spam emails}$$

$$P(x_j=1|y=1) = \phi_{j|y=1}$$

MLE:

$$\phi_y = \frac{\sum_{i=1}^m \mathbb{1}\{y^{(i)}=1\}}{m}$$

$$\phi_{j|y=0} = \frac{\sum_{i=1}^m \mathbb{1}\{x_j^{(i)}=1, y^{(i)}=0\}}{\sum_{i=1}^m \mathbb{1}\{y^{(i)}=0\}}$$

At prediction time:

$$P(y=1|x) = \frac{P(x|y=1)P(y=1)}{P(x|y=1)P(y=1) + P(x|y=0)P(y=0)}$$

$$\prod_{i=1}^{10000} P(x_i|y=1) = 0$$

그러면 만약 새 단어 $j=6317$ 이 포함된 이메일을 볼 때 시작한다면

$$P(x_{6317}=1|y=1) = \frac{0}{\# \text{ of } y=1}$$

이런 6317번째 단어는 본 적이 없으니 0으로 도둑 →

$$P(x_{6317}=0|y=0) = \frac{0}{\# \text{ of } y=0}$$

However, statistically it's just a bad idea to say that the chance of sth is zero

just because you haven't seen it. Also it makes 'divided by 0' issue.

Laplace smoothing

$$\text{꽃복숭이 4개라고 다진 상황, } P(x=1) = \frac{\#1's + 1}{\#1's + \#0's + 1} = \frac{0}{0+4} = 0 \rightarrow \frac{1}{1+5} = \frac{1}{6}$$

more generally:

$$x \in \{1, \dots, k\}, \text{ Estimate } P(x=j) = \frac{\sum_{i=1}^m \mathbb{1}\{x^{(i)}=j\} + 1}{m+1} \quad \leftarrow m+k \text{가 맞는지?}$$

When applying to Naive Bayes:

$$\phi_{j|y=0} = \frac{\sum_{i=1}^m \mathbb{1}\{x_j^{(i)}=1, y^{(i)}=0\} + 1}{\sum_{i=1}^m \mathbb{1}\{y^{(i)}=0\} + 2}$$

왜 1이 아니라 2?
x가 가능한 outcomes의 수 = k=2 (단어가 등장한다/안된다)

Discretize a continuous valued feature to discrete value features

$$x_i \in \{1, \dots, k\}$$

size	<400	400~600	600~1200	>1200
x_i	1	2	3	4

실제로는 10 buckets를 씀

$$P(x|y) = \prod_{j=1}^n P(x_j|y)$$

이전까지는 feature가 binary $\rightarrow$ Bernoulli NB
작은 feature가 k개의 discrete 값을 가짐 $\rightarrow$ 각 feature class에 대한 Multinomial NB 해야함

$P(x_j=k|y=c)$ 각 feature에 대한 multinomial 프로빈트 모델

$$\begin{aligned} \phi_{j=k|y=c} &= P(x_j=k|y=c) \\ &= \frac{\#(x_j=k, y=c) + 1}{\#(y=c) + k} \end{aligned}$$

이전까지는 Multivariate Bernoulli Event Model. 그러나 만약 같은 단어가 여러번 등장한다면?

New presentation: Multinomial Event Model (text에만 적용가능)

"Drugs! Buy drugs now!"

$$x \in \begin{bmatrix} 1600 \\ 800 \\ 1600 \\ 6200 \end{bmatrix} \in \mathbb{R}^{n_i}$$

$$p(y|x) = p(x|y) \cdot p(y)$$

Assume $\prod_{j=1}^n P(x_j|y)$

$x_j \in \{1, \dots, 10,000\}$ index (in dictionary)

n_i : The length of the email i 총 단어 수

parameters:

$$\phi_y = P(y=1)$$

$$\phi_{k|y=0} = P(x_j=k|y=0) \text{ Chance of word } k \text{ being } k \text{ when } y=0$$

$$\phi_{k|y=1} = P(x_j=k|y=1)$$

MLE:

$$\phi_{k|y=0} = \frac{\sum_{i=1}^m \mathbb{1}\{y^{(i)}=0\} \sum_{j=1}^{n_i} \mathbb{1}\{x_j^{(i)}=k\} + 1}{\sum_{i=1}^m \mathbb{1}\{y^{(i)}=0\} \cdot n_i + 10,000}$$

Laplace Smoothing

When to use Naive Bayes algorithm?

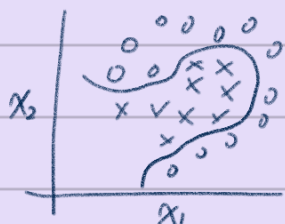
cons - not very competitive with other algorithms. For most cases, logistic algorithm will work better in terms of delivering higher accuracy.

pros - computationally very efficient / relatively quick to implement (small # of lines of code)
doesn't require an iterative gradient descent thing

+ SW package도 잘 되어 있음

(e.g. ϕ in regression)

Support Vector Machines "turn-key" algorithm - it does not have too many params to fiddle with



when you want to find the non-linear decision boundaries.

If, Logistic regression $\rightarrow$ feature $\Leftarrow$ high dimensional & augmented feature vector

$$\phi(x) = \begin{bmatrix} x_1 \\ x_2 \\ x_1^2 \\ x_2^2 \\ x_1 x_2 \\ \vdots \end{bmatrix}$$

바꾸면 ellipse 모양의 decision boundary 사용

더 확장된 decision boundary를 만들기 위한 feature preprocessing에
필요한 더 복잡

Optimal margin classifier (w separable case)

training data set can be linearly separated. $\leftarrow$ 이걸 기준으로 시작.

그후 kernels, inseparable case

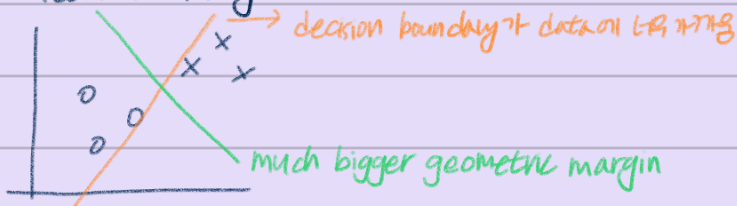
1. Functional margin how confidently and accurately do you classify an example.

Let's start with logistic regression:

$h_{\theta}(x) = g(\theta^T x)$ predict "1" if $\theta^T x \geq 0$ ($h_{\theta}(x) = g(\theta^T x) \geq 0.5$)
"0" otherwise

If $g^{(i)} = 1$, hope that $\theta^T x^{(i)} \gg 0$
If $g^{(i)} = 0$, hope that $\theta^T x^{(i)} \ll 0$ } this statement should be true.

2. Geometric margin



3. Notation

logistic regression 이걸 간단하게

Labels $y \in \{-1, +1\}$, Have h output values in $\{-1, +1\}$ $g(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ -1 & \text{otherwise} \end{cases}$

previously, $h_{\theta}(x) = g(\theta^T x)$ s.t. $x \in \mathbb{R}^n$, $x_0 = 1$

$$h_{w,b}(x) = g(\underbrace{w^T x + b}_{\text{Drop } x_0=1 \text{ convention}})$$

$$x \in \mathbb{R}^n, b \in \mathbb{R}$$

$$= \sum_{i=1}^n w_i x_i + b$$

$$\begin{bmatrix} \theta^0 \\ \theta^1 \\ \theta^2 \\ \vdots \end{bmatrix} \rightarrow b$$

4. Functional margin of ^(linear) hyperplane defined by (w, b) wrt $(x^{(i)}, y^{(i)})$

$$\hat{\gamma}^{(i)} = y^{(i)} (w^T x^{(i)} + b) \text{ for 1 training example}$$

If $y^{(i)} = 1$, want $w^T x^{(i)} + b \gg 0$ want $\hat{\gamma}^{(i)} \gg 0$
 If $y^{(i)} = -1$, want $w^T x^{(i)} + b \ll 0$

If $\hat{\gamma}^{(i)} > 0$, that means $h(x^{(i)}) = y^{(i)}$

functional margin to be the functional margin of worst training example

$$\hat{\gamma} = \min_i \hat{\gamma}^{(i)}$$

5. Normalize the parameters to rescale them to prevent cheating functional margin.

$$\|w\| = 1$$

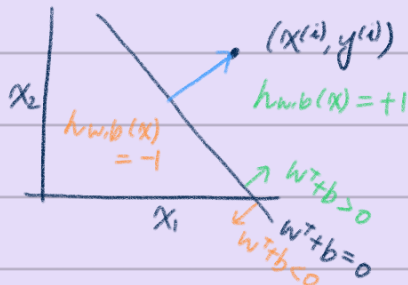
$$(w, b) = \left(\frac{w}{\|w\|}, \frac{b}{\|w\|} \right)$$

$$V = (V_1, V_2, \dots, V_n)$$

$$\|V\| = \sqrt{V_1^2 + V_2^2 + \dots + V_n^2} \text{ Euclidean length}$$

this doesn't change the classification result. $\|w\|$ can any number works.

6. Geometric margin of the classifier hyperplane defined by (w, b)



$$\gamma^{(i)} = \frac{y^{(i)} (w^T x^{(i)} + b)}{\|w\|}$$

$$= \frac{\hat{\gamma}^{(i)}}{\|w\|}$$

Geometric margin with training set $\gamma = \min_i \gamma^{(i)}$

7. Optimal margin classifier: choose w, b to maximize γ .

$$\max_{w, b} \gamma \text{ s.t. } \frac{y^{(i)} (w^T x^{(i)} + b)}{\|w\|} \geq \gamma \quad i = 1, \dots, m$$

$\Downarrow$

$$\min_{w, b} \|w\|^2 \text{ s.t. } y^{(i)} (w^T x^{(i)} + b) \geq 1$$